Midterm 2 content

Monday, March 9, 2026   4:00 PM

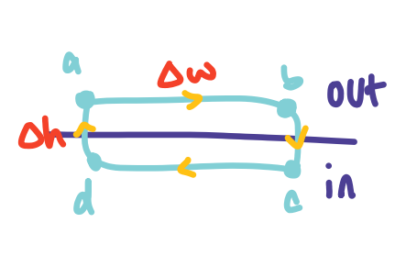
3-6 Static electric fields

materials: conductor, semiconductor, insulator (dielectric)

conductor: $\rho = 0$, $\vec{E} = 0$ inside

- under static conditions, $\vec{E}$ on surface everywhere normal
  - ↳ surface is equipotential

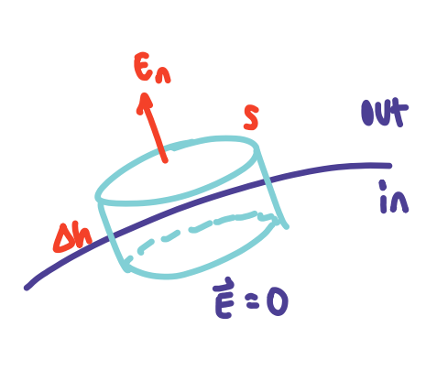
$$\oint_{abcda} \vec{E}\cdot d\vec{l} = E_t \Delta w = 0$$



$$\oint_{abcda} \vec{E}\cdot d\vec{l} = 0$$

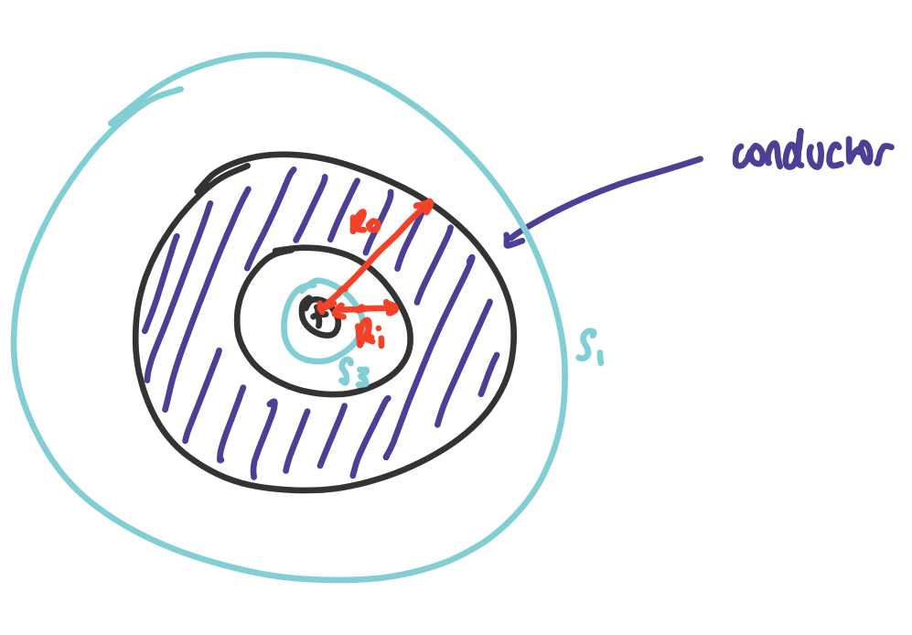
$$\underbrace{\int_a^b \vec{E}\cdot d\vec{l}}_{E_t \Delta w} + \underbrace{\int_b^c \vec{E}\cdot d\vec{l}}_{0\,(\Delta h \to 0)} + \underbrace{\int_c^d \vec{E}\cdot d\vec{l}}_{0\,(\text{inside})} + \underbrace{\int_d^a \vec{E}\cdot d\vec{l}}_{0\,(\Delta h \to 0)} = 0 \quad (\text{since } \oint_c \vec{E}\cdot d\vec{l} = 0)$$

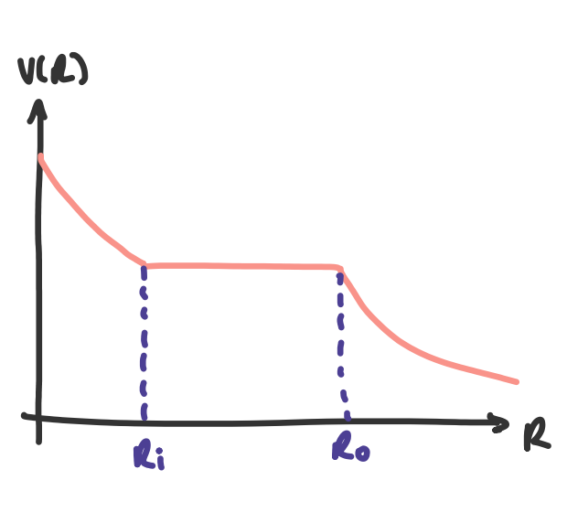
$$E_t \Delta w = 0 \Rightarrow \boxed{E_t = 0} \text{ (tangential)}$$



$$\oint_S \vec{E}\cdot d\vec{s} = \underbrace{\int_{top} \vec{E}\cdot d\vec{s}}_{E_n \Delta S} + \underbrace{\int_{bot} \vec{E}\cdot d\vec{s}}_{0\,(\vec{E}=0)} + \underbrace{\int_{side} \vec{E}\cdot d\vec{s}}_{0\,(\Delta h \to 0)}$$

$$E_n \Delta S = \frac{\rho_s \Delta S}{\varepsilon_0} \Rightarrow \boxed{E_n = \frac{\rho_s}{\varepsilon_0}} \text{ (normal)}$$





Gaussian surface S_1

$R > R_o$:

$$\oint_{S_1} \vec{E}\cdot d\vec{s} = E_{R1} 4\pi R^2 = \frac{Q}{\varepsilon_0}$$

$$\boxed{E_{R1} = \frac{Q}{4\pi\varepsilon_0 R^2}}$$

$$V_1 = -\int_\infty^R E_{R1}\, dR = \frac{Q}{4\pi\varepsilon_0 R}$$

inside conductor

$R_i < R < R_o$:

$$\boxed{E_{R2} = 0}$$

$$V_2 = V_1|_{R=R_o} = \frac{Q}{4\pi\varepsilon_0 R_o}$$

Gaussian surface S_3

$R < R_i$:

$$\oint_{S_3} \vec{E}\cdot d\vec{s} = E_{R3} 4\pi R^2 = \frac{Q}{\varepsilon_0} \quad (\text{same as } S_1)$$

$$\boxed{E_{R3} = \frac{Q}{4\pi\varepsilon_0 R^2}}$$

$$V_3 = -\int E_{R3}\, dR + \underset{\uparrow}{C} = \frac{Q}{4\pi\varepsilon_0 R} + C$$

need $V_3|_{R=R_i} = V_2 \Rightarrow \frac{Q}{4\pi\varepsilon_0 R_o} = \frac{Q}{4\pi\varepsilon_0 R_i} + C$

$$C = \frac{Q}{4\pi\varepsilon_0}\left(\frac{1}{R_o} - \frac{1}{R_i}\right)$$

$$V_3 = \frac{Q}{4\pi\varepsilon_0}\left(\frac{1}{R} + \frac{1}{R_o} - \frac{1}{R_i}\right)$$

3-7 dielectrics

$\rho_{ps} = \vec{P}\cdot\hat{a}_n$: component of polarization perpendicular to surface (surface charge)

$\rho_p = -\nabla\cdot\vec{P}$: leftover volume charge (unaligned dipoles) (volume charge)

$$\text{Total charge} = \oint_S \rho_{ps}\, ds + \int_V \rho_p\, dv$$
$$= \oint \vec{P}\cdot\hat{a}_n\, ds - \int_V \nabla\cdot\vec{P}\, dv = 0$$

generalized gauss law

$$\vec{\nabla}\cdot\vec{E} = \frac{\overset{\text{free}}{\rho_v} + \overset{\text{bound}}{\rho_p}}{\varepsilon_0}$$

$$\vec{\nabla}\cdot\underset{\text{electric flux density}}{\vec{D}} = \rho_v$$

$$\oint_S \vec{D}\cdot d\vec{s} = 0$$

$\vec{D}$ is charge per unit area of flux

$Q = \oint \vec{D}\cdot d\vec{A}$ is total free charge

$(Q = DA)$

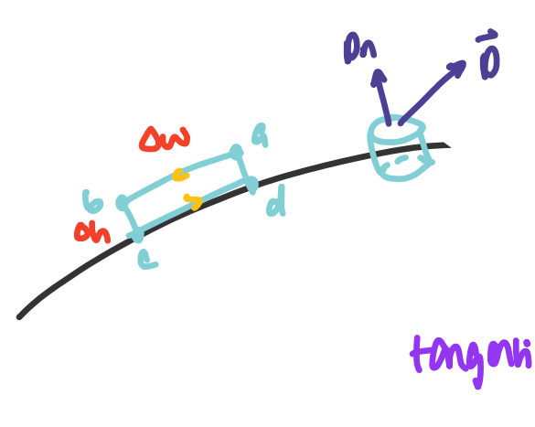
$$\vec{D} = \varepsilon_0 \overbrace{(1 + \underset{\text{dielectric susceptibility}}{\chi_e})}^{\varepsilon_r \text{ relative permittivity}} \vec{E}$$

$$\boxed{\vec{D} = \varepsilon_0\vec{E} + \vec{P}}$$

Fundamental postulates:

$$\begin{cases} \boxed{\nabla\cdot\vec{D} = \rho_v} \\ \boxed{\nabla\times\vec{E} = 0} \end{cases}$$

3-9 Boundary conditions



As $\Delta h \to 0$:

$$\oint_{abcda} \vec{E}\cdot d\vec{l} = E_{1t}\Delta w - E_{2t}\Delta w = 0$$

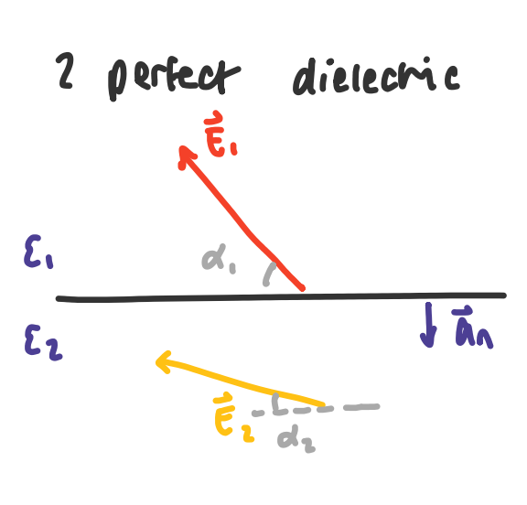
$$\boxed{E_{1t} = E_{2t}}$$

tangential component of $\vec{E}$ is continuous across interface

$$\int_S \vec{D}\cdot d\vec{s} = Q_{enc}$$

$$\boxed{\vec{D}_{2,n} - \vec{D}_{1,n} = \rho_s}$$ density of free charge @ interface

Ex. 2 perfect dielectric $(\rho_s = 0)$



$\rho_s = 0$

$D_{2,n} = D_{1,n} \Rightarrow \varepsilon_2 E_2 \sin\alpha_2 = \varepsilon_1 E_1 \sin\alpha_1$

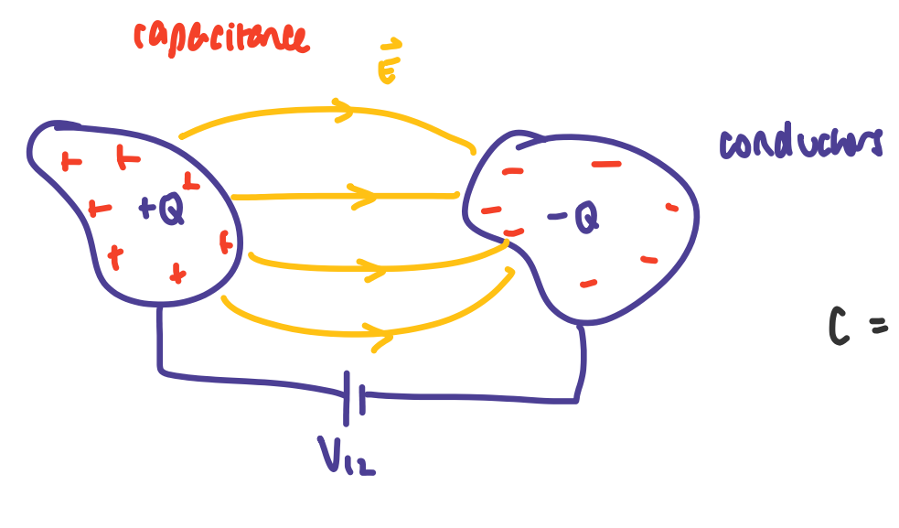
$E_{2,t} = E_{1,t} \Rightarrow E_2\cos\alpha_2 = E_1\cos\alpha_1$

$$\boxed{\varepsilon_2 \tan\alpha_2 = \varepsilon_1 \tan\alpha_1}$$

capacitance

Q always proportional to V

$$Q = \underset{\text{capacitance}}{C}\cdot V$$



$$C = \frac{Q}{V_{12}}$$

1. ① choose coord. sys.
2. ② assume charge $+Q$ (cond 1), $-Q$ (cond 2)
3. ③ find $\vec{E}$
4. ④ find $V_{12} = -\int \vec{E}\cdot d\vec{l}$
5. ⑤ $C = \frac{Q}{V_{12}}$

Ex. parallel plates

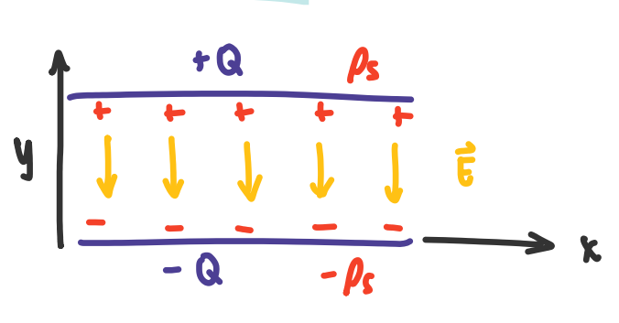


plate area: s

permittivity: ε

$$\left.\begin{aligned} \rho_s &= \frac{Q}{s} \\ -\rho_s &= -\frac{Q}{s} \end{aligned}\right\} \text{like } \vec{D}$$

$\vec{D} = \varepsilon\vec{E} + \vec{P}$

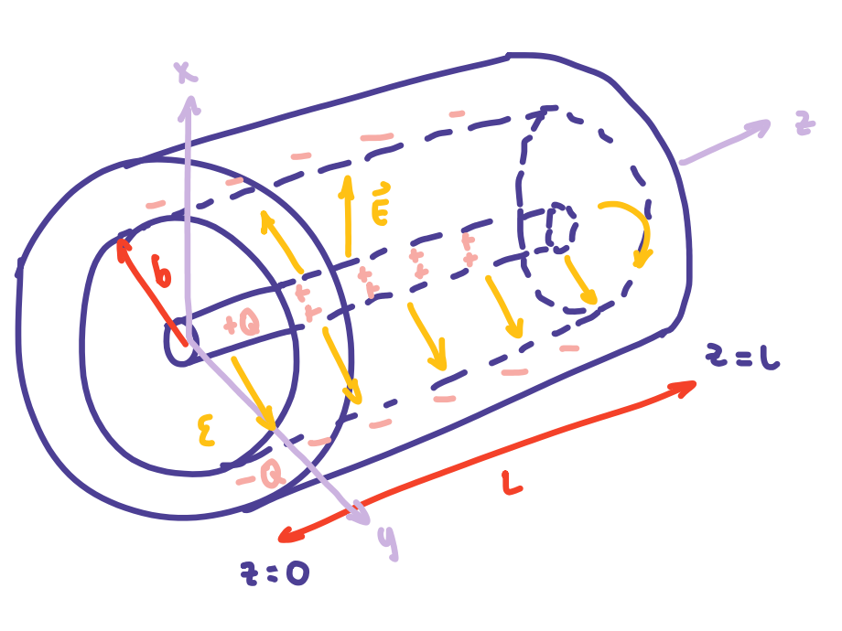
$$\vec{E} = \frac{\rho_s}{\varepsilon}(-\hat{a}_y) = -\frac{Q}{s\varepsilon}\hat{a}_y$$

$$V = -\int_{y=0}^d \vec{E}\cdot d\vec{l} = -\int_0^d -\frac{Q}{s\varepsilon}\underbrace{\vec{a}_y\cdot\vec{a}_y}_{1}\, dy = \frac{Q}{s\varepsilon} d$$

$$C = \frac{Q}{V} = Q\cdot\frac{s\varepsilon}{Qd}$$

$$\boxed{C = \varepsilon\frac{s}{d}}$$

Ex. cylinder



① coordinates: cylindrical

② assume $+Q$, $-Q$

③ $\vec{E} = E(r)\hat{a}_r = \frac{Q}{2\pi\varepsilon L r}\hat{a}_r$

↑ Gauss law: $\oint \vec{E}\cdot d\vec{A} = \frac{Q_{encl}}{\varepsilon_0}$

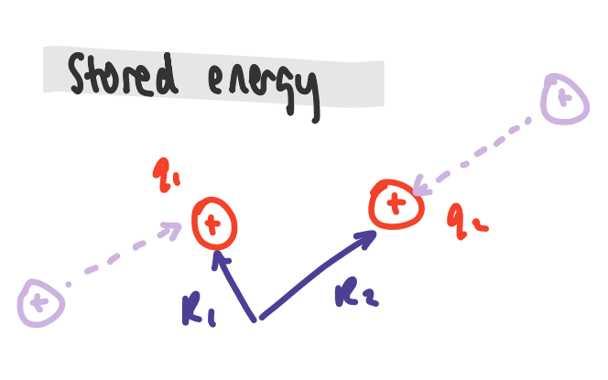
$E(r)\, 2\pi r L = \frac{Q}{\varepsilon_0}$

$E(r) = \frac{Q}{2\pi\varepsilon r L}$

④ $$V_{12} = -\int_{P_1}^{P_2} \vec{E}\cdot d\vec{l}$$
$$= -\int_{r=a}^{r=b} \frac{Q}{2\pi\varepsilon r L}\hat{a}_r\cdot\hat{a}_r\, dr$$
$$= \frac{Q}{2\pi\varepsilon L}\ln\frac{b}{a}$$

⑤ $$C = \frac{Q}{V} = \frac{2\pi\varepsilon L}{\ln\frac{b}{a}}$$

Stored energy



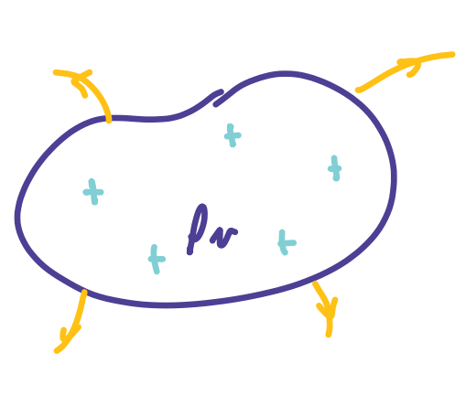
move q_1 to $\vec{R}_1$: $\boxed{W = 0}$ (no field from q_2 since far away)

move q_2 to $\vec{R}_2$: $\boxed{W = q_2\cdot V_1(\vec{R}_2)}$ (potential produced by q_1)

Total: $\boxed{W_e = q_2\cdot V_1(\vec{R}_2) = q_1\cdot V_2(\vec{R}_1)}$

Energy stored: $W_e = \frac{1}{2}\sum_{k=1}^{N} q_k\cdot V_{1k}(\vec{R}_k)$

continuous distribution

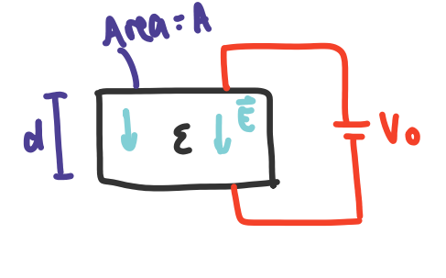


$$W_e = \underbrace{\frac{1}{2}\int_V \rho_v V\, dv'}_{\text{only where there's charge}} = \underbrace{\frac{1}{2}\int_{\mathbb{R}^3} \vec{D}\cdot\vec{E}\, dv'}_{\text{field over all space}} = \frac{1}{2}\int \varepsilon|\vec{E}|^2\, dv'$$

$$W_e = \frac{1}{2}CV^2 = \frac{1}{2}\int \varepsilon|\vec{E}|^2\, dv'$$

microscopic   macroscopic

Energy in a parallel plate capacitor



$$\vec{E} = \begin{cases} -\frac{V_0}{d}\hat{a}_z & \text{inside} \\ \sim 0 & \text{outside} \end{cases}$$

$$W_e = \frac{1}{2}\int_{\mathbb{R}^3} \varepsilon|\vec{E}|^2\, dv' = \frac{1}{2}\int_{\mathbb{R}^3} \varepsilon\frac{V_0^2}{d^2}\, dv'$$
$$= \frac{1}{2}\varepsilon\frac{V_0^2}{d^2}A\cdot d$$
$$= \frac{1}{2}\underset{\underset{C}{\uparrow}}{\boxed{\frac{\varepsilon A}{d}}}V_0^2$$

Gauss law: $\boxed{\nabla\cdot\vec{D} = \rho_v}$ ← $\vec{D} = -\varepsilon\vec{E} = -\varepsilon\nabla V$

Poisson eqn: $\boxed{\nabla\cdot(\varepsilon\underset{\text{potential}}{\nabla V}) = -\underset{\text{charge density}}{\rho_v}}$

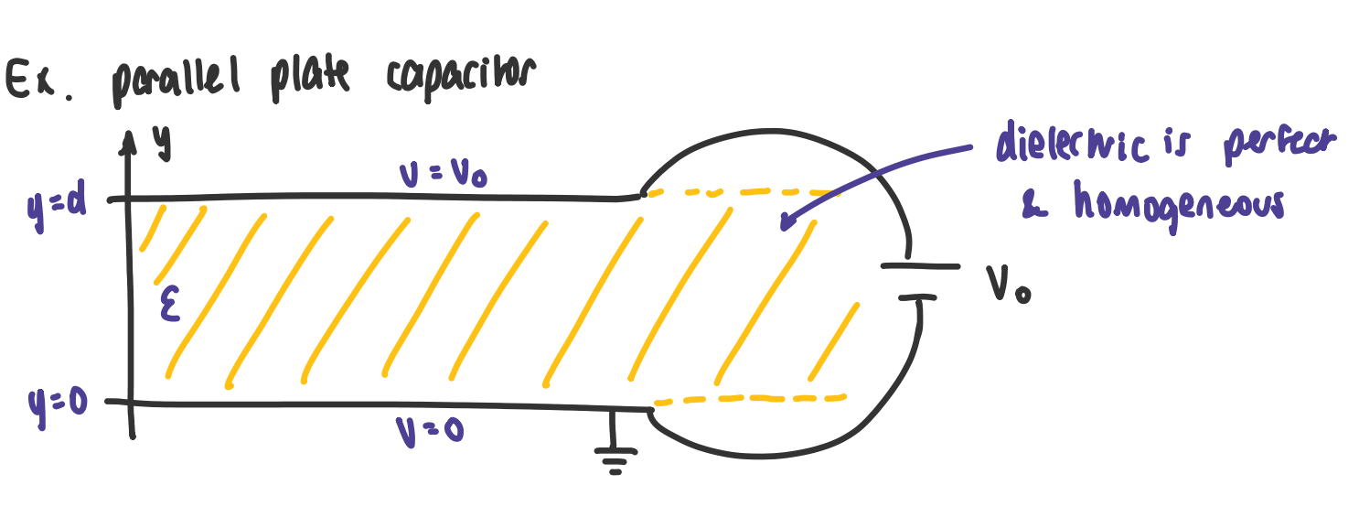
Solve boundary value problem

find V in volume v given:

given:
- V on boundary (V_0)
- ρ_v

$$\begin{cases} \nabla\cdot(\varepsilon\nabla V) = -\rho_v \\ V|_s = V_0 \end{cases}$$

Ex. parallel plate capacitor



$$\begin{cases} \nabla\cdot(\varepsilon\nabla V) = -\underbrace{\rho_v}_{0} \\ V(y=d) = V_0 \\ V(y=0) = 0 \end{cases}$$

$$\varepsilon\nabla\cdot\nabla V = -\rho_v$$
$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} \quad \text{Laplacian}$$

$$\begin{cases} \frac{\partial^2 V}{\partial y^2} = 0 \\ V(d) = V_0 \\ V(0) = 0 \end{cases} \Rightarrow V(y) = C_0 y + C_1$$

$$C_0 d + C_1 = V_0 \Rightarrow C_0 = \frac{V_0}{d},\quad C_1 = 0$$

$$V(y) = \frac{V_0}{d}y$$

current

$$dI = \frac{dq}{dt} = \rho\vec{u}\cdot d\vec{s}$$

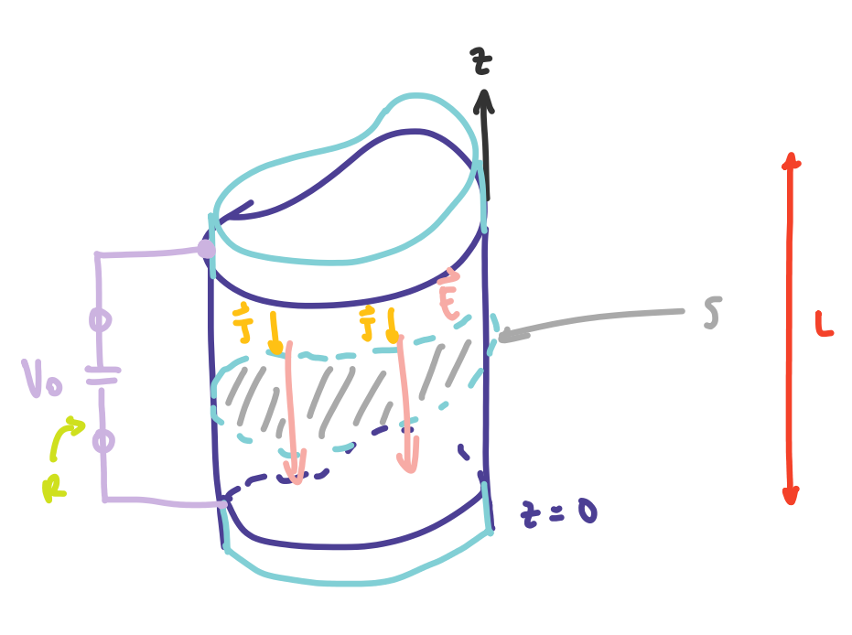
$$dq = \rho\, dv = \rho\, d\vec{l}\cdot d\vec{s} = \rho\underbrace{\vec{u}\, dt}_{d\vec{l}}\cdot d\vec{s}$$

Microscopic Ohm's Law: $\vec{J} = \sigma\vec{E}$

↑ current density $[A/m^2]$

$$I = \int_S \vec{J}\cdot d\vec{s}$$

R of straight wire



① coord. sys.

② apply voltage V_0

③ find $\vec{E}$

$$\vec{E} \sim (-\hat{a}_z)\cdot\frac{V_0}{L}$$

④ find $\vec{J} = \sigma\vec{E}$

$$\vec{J} = \frac{\sigma V_0}{L}(-\hat{a}_z)$$

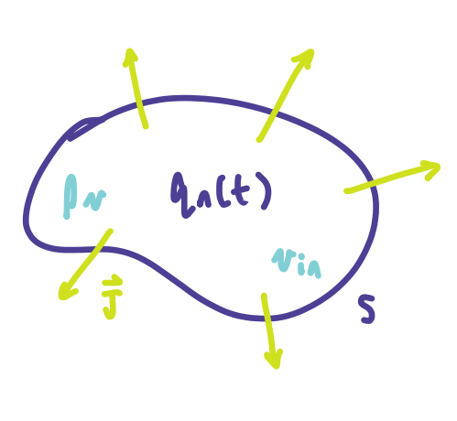
⑤ find $\vec{I} = \int_S \vec{J}\cdot d\vec{s}$

$$I = \int_S \vec{J}\cdot d\vec{s} = \int_S \frac{\sigma V_0}{L}\underbrace{(-\hat{a}_z)\cdot(-\hat{a}_z)}_{1}\, ds = \frac{\sigma V_0}{L}s$$

⑥ find $\underset{R = \frac{V_0}{I}}{R} = \frac{V_0}{\frac{\sigma V_0}{L}s}$

$$= \boxed{\frac{L}{\sigma s}}$$

Charge density



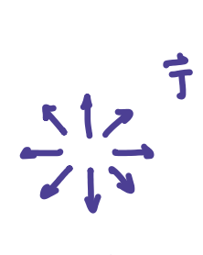
$$q_{in}(t) + q_{out}(t) = C$$

$$\frac{\partial q_{in}}{\partial t} + \frac{\partial q_{out}}{\partial t} = 0$$ ← current flowing through S

$$\boxed{\frac{\partial}{\partial t}\int_{v_{in}} \rho_v(t)\, dv + \int_S \vec{J}\cdot d\vec{s} = 0} \quad \text{continuity equation}$$

↓

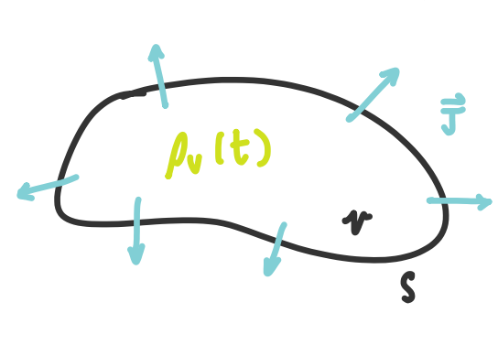
$$\frac{\partial\rho_v}{\partial t} + \vec{\nabla}\cdot\vec{J} = 0$$



$\vec{\nabla}\cdot\vec{J} > 0$

$\frac{\partial\rho_v}{\partial t} < 0$ ρ_v ↓ wrt. time

continuity equation = charge conservation



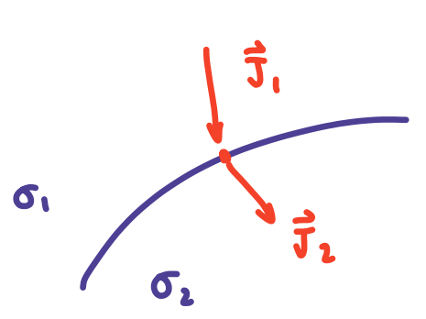
$$\underbrace{\int_S \vec{J}\cdot d\vec{s}}_{I} + \frac{\partial}{\partial t}\underbrace{\int_v \rho_v(t)\, dv}_{Q_{in}(t)} = 0$$

current flowing through S ($\frac{\text{charge}}{\text{time}}$)   change of charge inside

Joule's Law: $\boxed{P_d = \int_v \vec{E}\cdot\vec{J}\, dv}$

$$P_d = V\cdot I$$

Boundary conditions



$$\vec{J}_{1,n} = \vec{J}_{2,n}$$

$$\frac{J_{2,t}}{J_{1,t}} = \frac{\sigma_2}{\sigma_1}$$